Paediatric IV→Oral Antibiotic Switch Therapy (IVOST) Guideline → Can I switch my patient from IV to oral antibiotics?

Review need for IV antibiotics DAILY: review & document patient progress/the IVOST plan within 72 hours of IV antibiotic initiation

Are all of the following IVOST criteria met?

- ✓**CLINICAL IMPROVEMENT** in signs of infection e.g. temperature ≤ 37.9°C , reduction in the PEWS score, improving SEPSIS
- ✓**ORAL ROUTE** **is available reliably** (eating/drinking and no concerns regarding absorption)
- ✓**UNCOMPLICATED INFECTION** (specialist advice not required prior to IVOST). Certain infections need prolonged IV e.g. CNS infection; Cystic Fibrosis; Endocarditis; Bone/joint infection; Undrainable deep abscess; Bacteraemia.

CRP does NOT reflect severity of illness or the need for IV antibiotics & may remain elevated as the infection improves.

DO NOT use CRP in isolation to assess IVOST suitability. Most infections require ≤ 7 days TOTAL (IV + oral) therapy. Record the intended duration on HEPMA/Kardex

YES

Can you STOP THERAPY? If antibiotic therapy is still required → SWITCH TO ORAL.
First, check the MICROBIOLOGY results; can you NARROW THE ANTIBIOTIC SPECTRUM based on cultures?

If the patient is being treated in line with the Empiric Infection Management Guidelines AND there is NO POSITIVE MICROBIOLOGY to guide the change then switch to oral as outlined below;

DIAGNOSIS❖	EMPIRIC ORAL SWITCH■ 1ST LINE	2nd LINE/PENICILLIN ALLERGY	TOTAL duration (IV + PO)
Resolving sepsis and source unknown	**Discuss with micro/ID**	**Discuss with micro/ID**	**As per micro/ID**
Community-acquired pneumonia (CAP)	Amoxicillin	Azithromycin	5 days. Give 3 days PO Azithromycin
Aspiration pneumonia	Amoxicillin	Clindamycin	5 - 10 days based on clinical improvement
Severe pneumonia/pnemonia complicating influenza	Co-amoxiclav	Discuss with Micro/ID	Discuss with Micro/ID
Acute mastoiditis	Co-amoxiclav	Clindamycin	14 days
Intra-abdominal sepsis	Co-amoxiclav	Ciprofloxacin & metronidazole	7 days if source control, if not discuss with Micro/ID
Upper tract UTI/Pyelonephritis	Cefalexin	Ciprofloxacin	7-10 days
Septic arthritis or Osteomyelitis	Co-amoxiclav (If ≥6yrs considser Flucloxacillin capsules)	Clindamycin	3 weeks for septic arthritis and 4-6 weeks for osteomyelitis
Cellulitis	Flucloxacillin	Clindamycin	7 days
Orbital/Peri-orbital cellulitis	Co-amoxiclav	Clindamycin	7-10 days

NO

Check the microbiology results. Can you NARROW THE SPECTRUM of IV therapy?

Is the patient on IV gentamicin?

DO NOT continue IV gentamicin for longer than 4 days (except on the advice of an infection specialist - Microbiology/ID). Patients prescribed IV gentamicin for 4 days should have a senior review of their diagnosis, microbiology and clinical progress;

1. Is Gram-negative cover still required? If not, stop gentamicin.
2. Is there any positive microbiology? If so rationalise.
3. If IV therapy & Gram-negative cover are still required and there is no positive microbiology, discuss with Microbiology/ID.
4. If gentamicin is required for longer (e.g. endocarditis) seek Microbiology/ID advice & monitor for signs of renal & oto/vestibular toxicity.

❖Consult an infection specialist (microbiology or infectious diseases) on the switch options for the complicated infections listed above OR where empiric IV antibiotic therapy differed from guideline on specialist advice.

■ Consult BNFC for dosing advice

Antimicrobial	Oral bioavailability
Metronidazole	>90%
Clindamycin	>90%
Ciprofloxacin	80%

Greater Glasgow & Clyde Antimicrobial Utilisation Committee, November 2024. Review November 2027.